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import numpy as np

# Sample dataset
english = [
    "hello", "thank you", "good morning", "good night", "how are you", "i am fine", "what is your name",
    "my name is", "nice to meet you", "see you soon", "goodbye", "please", "sorry", "yes", "no",
    "i love you", "help me", "stop", "go", "come here", "sit down", "stand up", "open the door", "close the door",
    "what time is it", "i am hungry", "i am thirsty", "where is the toilet", "i am tired", "i am happy",
    "i am sad", "do you speak english", "i don't understand", "repeat please", "can you help me",
    "how much is this", "i want this", "i don't want this", "call the police", "it's urgent",
    "i am lost", "where am i", "turn left", "turn right", "go straight", "stop here", "be careful",
    "happy birthday", "congratulations", "merry christmas", "happy new year", "best wishes",
    "good luck", "have a nice day", "take care", "i miss you", "what happened", "i am sorry",
    "don't worry", "everything is fine", "enjoy", "let's go", "wait for me", "come with me",
    "where are you going", "i am going home", "do you need help", "thank you very much",
    "see you later", "i understand", "good job", "well done", "nice work", "beautiful",
    "delicious", "what is this", "this is mine", "this is yours", "who are you", "what do you do",
    "where are you from", "i am from india", "do you like it", "i like it", "i don't like it",
    "it's okay", "very good", "not good", "maybe", "of course", "never", "sometimes", "always",
    "let's eat", "i am coming", "wait", "now", "later", "today", "tomorrow", "yesterday"
]

hindi = [
    "नमस्ते", "धन्यवाद", "सुप्रभात", "शुभ रात्रि", "आप कैसे हैं", "मैं ठीक हूँ", "आपका नाम क्या है",
    "मेरा नाम है", "आपसे मिलकर खुशी हुई", "जल्द मिलते हैं", "अलविदा", "कृपया", "माफ कीजिए", "हाँ", "नहीं",
    "मैं तुमसे प्यार करता हूँ", "मेरी मदद करो", "रुको", "जाओ", "यहाँ आओ", "बैठ जाओ", "खड़े हो जाओ", "दरवाजा खोलो", "दरवाजा बंद करो",
    "क्या समय हुआ है", "मुझे भूख लगी है", "मुझे प्यास लगी है", "शौचालय कहाँ है", "मैं थक गया हूँ", "मैं खुश हूँ",
    "मैं दुखी हूँ", "क्या आप अंग्रेजी बोलते हैं", "मैं नहीं समझा", "कृपया दोहराएं", "क्या आप मेरी मदद कर सकते हैं",
    "यह कितने का है", "मुझे यह चाहिए", "मुझे यह नहीं चाहिए", "पुलिस को बुलाओ", "यह जरूरी है",
    "मैं खो गया हूँ", "मैं कहाँ हूँ", "बाएँ मुड़ो", "दाएँ मुड़ो", "सीधे जाओ", "यहाँ रुकें", "सावधान रहें",
    "जन्मदिन मुबारक हो", "बधाई हो", "क्रिसमस की शुभकामनाएं", "नव वर्ष की शुभकामनाएं", "शुभकामनाएं",
    "शुभकामनाएं", "आपका दिन शुभ हो", "ख्वाल रखना", "मुझे तुम्हारी याद आती है", "क्या हुआ", "मुझे खेद है",
    "चिंता मत करो", "सब कुछ ठीक है", "आनंद लो", "चलो चलते हैं", "मेरे लिए प्रतीक्षा करें", "मेरे साथ आओ",
    "आप कहाँ जा रहे हैं", "मैं घर जा रहा हूँ", "क्या आपको मदद चाहिए", "बहुत बहुत धन्यवाद",
    "फिर मिलेंगे", "मैं समझता हूँ", "शाबाश", "बहुत अच्छा", "अच्छा काम", "सुंदर",
    "स्वादिष्ट", "यह क्या है", "यह मेरा है", "यह तुम्हारा है", "आप कौन हैं", "आप क्या करते हैं",
    "आप कहाँ से हैं", "मैं भारत से हूँ", "क्या आपको यह पसंद है", "मुझे यह पसंद है", "मुझे यह पसंद नहीं है",
    "ठीक है", "बहुत अच्छा", "अच्छा नहीं", "शायद", "बिल्कुल", "कभी नहीं", "कभी-कभी", "हमेशा",
    "चलिए खाएं", "मैं आ रहा हूँ", "प्रतीक्षा करें", "अब", "बाद में", "आज", "कल", "कल (बीता हुआ)"
]

# Create character sets
input_chars = ['<PAD>'] + sorted(set(''.join(english)))
target_chars = ['<SOS>', '<EOS>', '<PAD>'] + sorted(set(''.join(hindi)))

# Create mappings
input_char_to_idx = {c: i for i, c in enumerate(input_chars)}
target_char_to_idx = {c: i for i, c in enumerate(target_chars)}
idx_to_target_char = {i: c for c, i in target_char_to_idx.items()}

# Helper: One-hot encoding
def one_hot_encode(sentence, char_to_idx, vocab_size, max_len):
    vec = np.zeros((max_len, vocab_size))
    for i, char in enumerate(sentence[:max_len]):
        if char in char_to_idx:
            vec[i][char_to_idx[char]] = 1
    for i in range(len(sentence), max_len):
        vec[i][char_to_idx['<PAD>']] = 1
    return vec

# Configuration
max_input_len = 10
max_output_len = 12

# Prepare training data
X = np.array([one_hot_encode(sent, input_char_to_idx, len(input_chars), max_input_len) for sent in english])
y = np.array([one_hot_encode('<SOS>' + sent + '<EOS>', target_char_to_idx, len(target_chars), max_output_len) for sent in hindi])

input_size = len(input_chars)
hidden_size = 64
output_size = len(target_chars)

# Initialize weights
Wxh = np.random.randn(hidden_size, input_size) * 0.01
Whh = np.random.randn(hidden_size, hidden_size) * 0.01
Why = np.random.randn(output_size, hidden_size) * 0.01

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bh = np.zeros((hidden_size, 1))
by = np.zeros((output_size, 1))

# Softmax function
def softmax(x):
    e_x = np.exp(x - np.max(x))
    return e_x / e_x.sum(axis=0)

# Training loop
for epoch in range(500):
    total_loss = 0
    for i in range(len(X)):
        h = np.zeros((hidden_size, 1))
        for t in range(max_input_len):
            x = X[i][t].reshape(-1, 1)
            h = np.tanh(np.dot(Wxh, x) + np.dot(Whh, h) + bh)

        loss = 0
        for t in range(max_output_len):
            y_pred = softmax(np.dot(Why, h) + by)
            target = y[i][t].reshape(-1, 1)
            loss += -np.sum(target * np.log(y_pred + 1e-9)) # avoid log(0)

        total_loss += loss
    if epoch % 100 == 0:
        print(f"Epoch {epoch}, Loss: {total_loss:.4f}")

# Translation function
def translate(inp):
    inp_vec = one_hot_encode(inp, input_char_to_idx, len(input_chars), max_input_len)
    h = np.zeros((hidden_size, 1))
    for t in range(max_input_len):
        x = inp_vec[t].reshape(-1, 1)
        h = np.tanh(np.dot(Wxh, x) + np.dot(Whh, h) + bh)

    result = ""
    for _ in range(max_output_len):
        y_pred = softmax(np.dot(Why, h) + by)
        idx = np.argmax(y_pred)
        char = idx_to_target_char[idx]
        if char == '<EOS>':
            break
        if char not in ['<PAD>', '<SOS>']:
            result += char
    return result

# Test on unseen input
print("=== Unseen Input Test ===")
unseen_inputs = ["good job", "nice", "do work"]
for sentence in unseen_inputs:
    print(f"English: {sentence}")
    print(f"Hindi: {translate(sentence)}\n")

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→ Epoch 0, Loss: 2656.8469
Epoch 100, Loss: 2656.8469
Epoch 200, Loss: 2656.8469
Epoch 300, Loss: 2656.8469
Epoch 400, Loss: 2656.8469
=== Unseen Input Test ===
English: good job
Hindi: सससससससससससस

English: nice
Hindi: सससससससससससस

English: do work
Hindi: सससससससससससस

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